MODIFICATIONS OF A COMPLETE 2-DIMENSIONAL LOCAL DOMAIN ARE PROJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. Thin card at the H2 floor: the reduction direction (criterion says 'suffices to check the fibre') is real logic, not a bare pointer.

Allowed standard prerequisites (editorial).

- Projectivity criterion over a complete base. More on Morphisms of Spaces, lemma projective-over-complete (cross-chapter, named). Reduces projectivity of a modification over the complete base to the special fibre having dimension at most 1.
- Special fibre dimension bound. lemma-dimension-special-fibre (carded L26825). Supplies dim <= dim(A) - 1 = 1 for the special fibre of a modification.

Source reference. Original lemma 0AYJ: sources/stacks-spaces-resolve/pinned-original-source.tex, source lines 322–322. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0AE7: focused-seeds/geometry/S9-spres-modification-of-dim-2-is-projective-over-complete/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *If $(A, \mathfrak{m}, \kappa)$ is a complete Noetherian local domain of dimension 2, then every modification of* $\mathrm{Spec}(A)$ *is projective over A.*

Proof. By More on Morphisms of Spaces, Lemma 0AE7 it suffices to show that the special fibre of any modification X of $\mathrm{Spec}(A)$ has dimension ≤ 1. This follows from Lemma 0AYJ. □

Date: stacks-project:spaces-resolve.tex:local-snapshot-20261012; GNU FDL 1.2 or later.